

RFID BASED AUTHENTICATION SYSTEM Using Arduino

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The radio frequency identification (RFID) system used for identification and tracking of objects makes use of an RFID reader and a tag. The main parts of the device include the transceiver, antenna, and decoder.

The transceiver can function as either a transmitter or a receiver. It includes an oscillator to generate a continuous signal, which is modulated to a required frequency and then transmitted into the air through the antenna.

The antenna converts the electrical signal into an electromagnetic signal that is efficiently propagated through the air.

When an RF signal is detected at the antenna from a tag, the decoder aids in retrieving the data.

The importance of RFID systems is increasing day by day due to their powerful capabilities in automatic identification, localisation, and access control. However, RFID techniques are plagued by security and privacy issues due to the underlying wireless communication channel. To address this, we propose an efficient authentication scheme that utilises pseudorandom

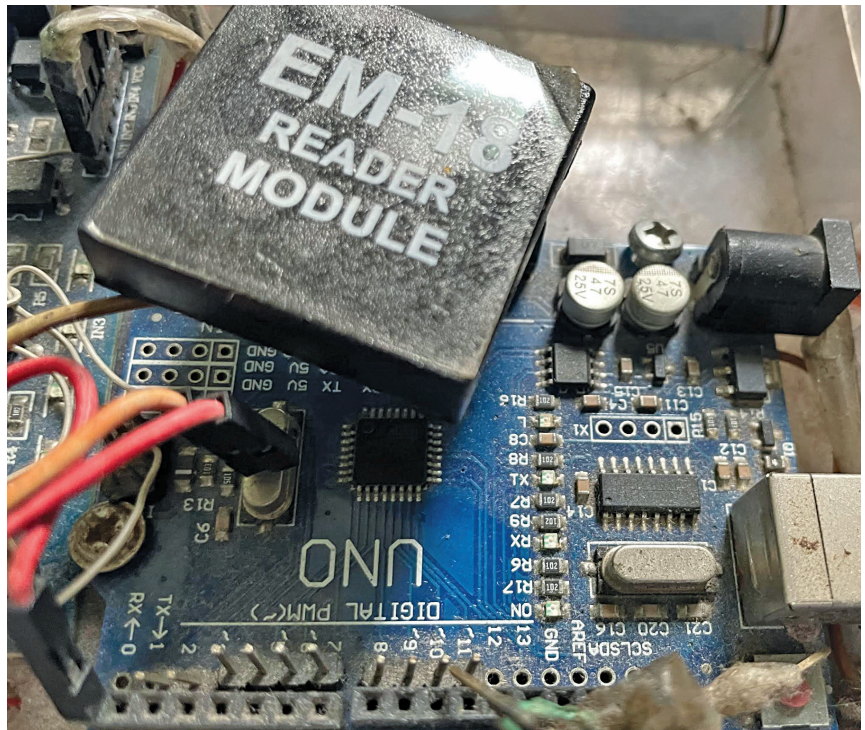


Fig. 1: Prototype tested in EFY Lab

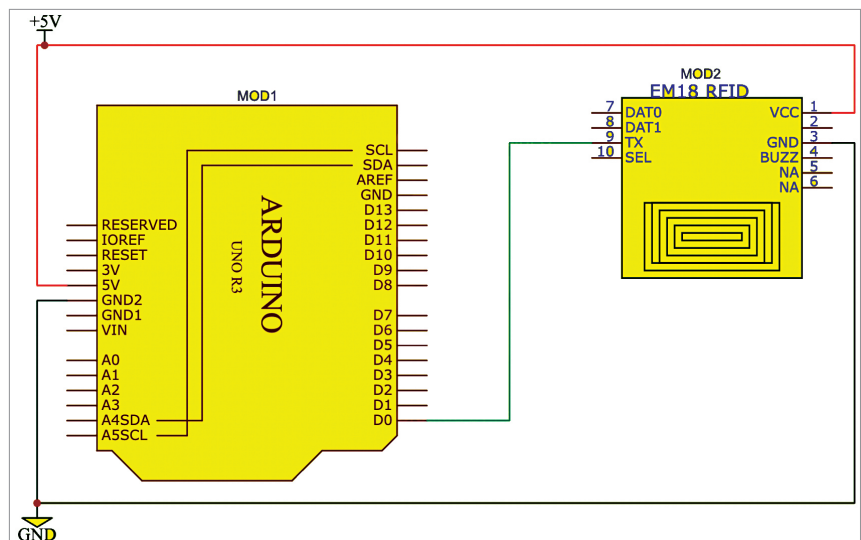


Fig. 2: Circuit diagram

BILL OF MATERIALS

Components	Quantity
Arduino Nano/Uno (MOD1)	1
EM18 RFID reader module (MOD2)	1
Jumper wires	4
RFID card/Tag for testing	1
5V adaptor	1